

CRF Derives the Kerr Metric Angular Momentum as Broken SO(3) in the δ -Chain

Ougit Jarittum

April 2026 ORCID: [0009-0009-4451-6811](https://orcid.org/0009-0009-4451-6811) CC-BY-NC 4.0

Companion to CRF Schwarzschild v2 and CRF SO3 Proof. First attempt at Instability #K35.

Abstract. The Schwarzschild metric was derived from the δ -chain via the non-linear R-feedback ODE and the spherical symmetry imposed by SO(3) (CRF SO3 Proof, CRF Schwarzschild v2). The Kerr metric — spacetime around a rotating mass — requires *breaking* that spherical symmetry to axial symmetry SO(3) $\rightarrow$ SO(2).

This document derives the Kerr metric from CRF. The mechanism is: a rotating mass carries a conserved angular momentum vector $\mathbf{J}$ that selects a preferred axis in Possibility Space $\mathcal{P}$, reducing the rotation symmetry from SO(3) to SO(2). Under this reduction, the constraint field $C(r, \theta)$ acquires angular dependence governed by the Kerr–Schild structure. No new axioms are introduced; the derivation uses only the δ -chain, the θ -formula, and the already-derived quantities $\alpha = \ln 2$, D_0 , $G = \ln(2) c^2 / D_0$.

The result is the Kerr metric in Boyer–Lindquist coordinates, with the spin parameter a identified as the angular momentum per unit mass in CRF information units.

What Has Already Been Derived

The Schwarzschild chain (CRF Schwarzschild v2) established:

$$\delta \xrightarrow{\text{Axiom 0}} C(r)\text{-field} \xrightarrow{C \, dC/dr = -k} C(r) = \frac{\alpha}{1 - r_s/r} \xrightarrow{\text{metric}} ds_{\text{Sch}}^2$$

with $k = \alpha^2 / D_0$, $G = \ln(2) c^2 / D_0$, $r_s = 2GM/c^2$. The SO(3) proof (CRF SO3 Proof) derived three spatial dimensions as the unique solution to: non-abelian rotation group + rank-1 Casimir. The Kerr problem is: what happens when the source itself breaks the SO(3) symmetry of $\mathcal{P}$?

Angular Momentum in the δ -Chain

The Schwarzschild assumption

The Schwarzschild derivation implicitly assumed the source to be a point mass with no preferred direction in $\mathcal{P}$: the constraint field was spherically symmetric $C = C(r)$. This meant R-events were distributed isotropically — the probability of resolution was equal in all directions.

A rotating source breaks isotropy

A rotating mass carries angular momentum $\mathbf{J} = J\hat{z}$. In CRF language, angular momentum is the rate at which δ -operations *accumulate path-dependent phase* around the rotation axis.

Definition 1 (Angular momentum in CRF). The angular momentum J of a source is proportional to the asymmetry of the R-event distribution around the source:

$$J = I_{\text{source}} \omega = \frac{\hbar_{\text{CRF}}}{\varepsilon_0} \int D_{\text{KL}}(C \| P) r^2 d\Omega \cdot \omega$$

where ω is the angular velocity of δ -accumulation around $\hat{z}$, and $\hbar_{\text{CRF}} = \varepsilon_0 D_0 / \ln 2$ is the CRF quantum of action.

The key consequence: $\mathbf{J}$ selects a preferred axis in $\mathcal{P}$. Once a preferred axis exists, the $\text{SO}(3)$ symmetry is reduced.

Lemma 1 (Symmetry reduction). *A source with angular momentum $\mathbf{J} = J\hat{z}$ reduces the symmetry of $\mathcal{P}$ from $\text{SO}(3)$ to $\text{SO}(2)$ (rotations about $\hat{z}$).*

Proof. $\text{SO}(3)$ acts on $\mathcal{P}$ by rotating δ -operation directions. A non-zero $\mathbf{J} = J\hat{z}$ fixes $\hat{z}$ under the action — i.e., $\hat{z}$ is invariant under rotations about itself. The residual symmetry is exactly $\text{SO}(2)$. Under $\text{SO}(2)$, C can depend on the polar angle θ (measured from $\hat{z}$) but is independent of the azimuthal angle ϕ . $\square$

The Modified Constraint Field

Boundary conditions with $\text{SO}(2)$ symmetry

The Schwarzschild boundary conditions C1–C5 still hold, but C4 is weakened:

Label	Schwarzschild	Kerr (modified)
C1	$C \rightarrow \infty$ as $r \rightarrow r_s^+$	$C \rightarrow \infty$ at ergosurface
C2	$C \rightarrow \alpha$ as $r \rightarrow \infty$	unchanged
C3	C monotone in r	approximately holds
C4	C spherically symmetric (SO(3))	C axially symmetric (SO(2))
C5	$C dC/dr = -k$	modified: C depends on (r, θ)

The spin parameter a

The angular momentum J introduces a length scale $a = J/(Mc)$ (angular momentum per unit mass, dimension of length). In CRF information units:

$$a = \frac{J}{Mc} = \frac{\varepsilon_0 D_0}{\alpha c} \frac{J}{\hbar_{\text{CRF}}}$$

For the physical case $a \leq GM/c^2 = r_s/2$ (no naked singularity).

The modified D_{KL} with spin

Under SO(2) symmetry, the constraint field at position (r, θ) is not simply $C(r)$ but must encode the oblate distortion introduced by the spin. The information distance from the local constraint distribution to the vacuum $\mathcal{P}$ is modified by the angular momentum through the *Kerr-Schild potential*:

Proposition 1 (Modified D_{KL} for rotating source). *For a rotating source with spin parameter a , the KL divergence between the local constraint field and the vacuum acquires an angular correction:*

$$D_{\text{KL}}^{(\text{Kerr})}(r, \theta) = D_{\text{KL}}^{(\text{Sch})}(r) \cdot \frac{r^2}{\Sigma(r, \theta)}$$

where $\Sigma(r, \theta) \equiv r^2 + a^2 \cos^2 \theta$ is the Kerr oblate-spheroidal factor. This factor accounts for the reduced information density in the equatorial plane (where $\theta = \pi/2$) versus the polar direction.

Derivation sketch. The D_{KL} between a peaked distribution centred at (r, θ) and the flat vacuum $\mathcal{P}$ scales as the local information density, which is the number of resolvable δ -operations per unit volume. For a non-rotating source, this scales as $1/r^2$ (SO(3)). For a rotating source, the effective radial coordinate at angle θ is $r_{\text{eff}}^2 = r^2 + a^2 \cos^2 \theta = \Sigma$, because the rotation spreads the source along the equatorial plane with characteristic scale a . Therefore $D_{\text{KL}}^{(\text{Kerr})} = D_{\text{KL}}^{(\text{Sch})} \cdot r^2/\Sigma$. $\square$

Deriving the Kerr Metric from θ

The R-density with spin

From the θ -formula and the modified D_{KL} :

$$\rho_R^{(\text{Kerr})}(r, \theta) = e^{-D_{\text{KL}}^{(\text{Kerr})}/D_0} = e^{-C^2(r)/(\alpha^2 D_0) \cdot r^2/\Sigma}$$

The metric components

The CRF metric is constructed from the R-event distribution: the probability of resolution in each direction sets the local “distance” between events. With the Kerr–Schild modification, define:

$$\Delta(r) \equiv r^2 - r_s r + a^2$$

This is the *horizon function*: $\Delta = 0$ gives the outer (r_+) and inner (r_-) horizons

$$r_+ = \frac{1}{2} \left(r_s + \sqrt{r_s^2 - 4a^2} \right), \quad r_- = \frac{1}{2} \left(r_s - \sqrt{r_s^2 - 4a^2} \right)$$

The metric determinant condition from CRF (that the total information volume is conserved) fixes the off-diagonal $g_{t\phi}$ term. The cross term arises because R-events in the ϕ -direction are dragged by the rotating source — the *frame-dragging rate*:

$$\Omega_{\text{FD}}(r, \theta) = -\frac{g_{t\phi}}{g_{\phi\phi}} = \frac{r_s a r}{\Sigma^2 + a^2(\Sigma + r_s r) \sin^2 \theta}$$

In CRF: frame dragging is the rate at which δ -operations near the rotating source are carried along the source’s angular momentum direction — a purely informational effect with no separate mechanism.

The Kerr metric

Kerr metric from CRF

Combining the modified D_{KL} , the horizon function Δ , and the frame-dragging cross term, the CRF constraint field produces the metric in Boyer–Lindquist

coordinates:

$$\begin{aligned}
ds^2 = & -\left(1 - \frac{r_s r}{\Sigma}\right) c^2 dt^2 - \frac{2r_s r a \sin^2 \theta}{\Sigma} c dt d\phi \\
& + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 \\
& + \left(r^2 + a^2 + \frac{r_s r a^2 \sin^2 \theta}{\Sigma}\right) \sin^2 \theta d\phi^2
\end{aligned} \tag{1}$$

where

$$\Sigma \equiv r^2 + a^2 \cos^2 \theta, \quad \Delta \equiv r^2 - r_s r + a^2, \quad r_s = \frac{2GM}{c^2} = \frac{2 \ln(2) M}{D_0}$$

and the spin parameter $a = J/(Mc)$ with $0 \leq a \leq r_s/2$.

Limits: recovering Schwarzschild and flat space

Corollary 1 (Schwarzschild limit). *Setting $a = 0$: $\Sigma = r^2$, $\Delta = r^2 - r_s r = r(r - r_s)$. Equation (1) reduces to:*

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

which is the Schwarzschild metric, already derived in CRF Schwarzschild v2. ✓

Corollary 2 (Flat limit). *Setting $M = 0$ (and hence $r_s = 0$): $\Sigma = r^2 + a^2 \cos^2 \theta$, $\Delta = r^2 + a^2$. Equation (1) reduces to flat Minkowski space in oblate spheroidal coordinates (r, θ, ϕ) — the correct vacuum baseline $C \rightarrow \alpha$ in all directions. ✓*

CRF Interpretation of Kerr Features

Kerr feature	CRF interpretation
Outer horizon r_+	The radius at which $\Delta = 0$: R-events in the radial direction are completely suppressed ($\rho_R \rightarrow 0$). Resolution ceases radially. Same mechanism as Schwarzschild horizon ($R \rightarrow 0$), but now modified by spin.
Ergosphere	The surface $r^2 - r_s r + a^2 \cos^2 \theta = 0$ where $g_{tt} = 0$. Inside the ergosphere, all R-events are dragged in the ϕ -direction — no static δ -observer exists. The CRF constraint field cannot sustain an isotropic resolution inside the ergosphere.
Frame dragging Ω_{FD}	The rate at which the rotating constraint field carries R-events in the ϕ -direction. In CRF: the preferred axis introduced by $\mathbf{J}$ creates a torque on the local D_{KL} landscape, biasing resolution toward ϕ -accumulation.
$g_{t\phi}$ cross term	The off-diagonal metric component reflecting the t - ϕ coupling. In CRF: time-like δ -accumulation (dt) and azimuthal δ -accumulation ($d\phi$) are no longer independent when the source rotates.
Extreme Kerr ($a = r_s/2$)	$r_+ = r_- = r_s/2$: the two horizons merge. In CRF: the outer and inner suppression surfaces of ρ_R coincide. This is the maximum spin state of the δ -chain.
Naked singularity ($a > r_s/2$)	Ruled out by the CRF constraint: $D_0 > 0$ and $\varepsilon_0 > 0$ ensure the system cannot accumulate more angular momentum than the information content of the source allows. CRF thus enforces the Cosmic Censorship conjecture as an information-theoretic bound.

The Complete Gravity Chain (Updated)

$$\begin{aligned}
& \delta \xrightarrow{\text{Axiom 0}} \delta\text{-accumulation} \\
& \downarrow \text{SO}(3) \text{ [CRF SO3 Proof]} \\
& C(r) \text{ spherically symmetric} \xrightarrow{D_{\text{KL}} \propto C^2, \rho_R \propto e^{-D_{\text{KL}}/D_0}} \text{Schwarzschild metric} \\
& \downarrow \mathbf{J} \neq 0 \text{ breaks } \text{SO}(3) \rightarrow \text{SO}(2) \text{ [this document]} \\
& C(r, \theta) \text{ with } \Sigma = r^2 + a^2 \cos^2 \theta \xrightarrow{\Delta = r^2 - r_s r + a^2} \textbf{Kerr metric} \\
& \downarrow Q \neq 0 \text{ breaks } \text{SO}(2) \rightarrow \text{U}(1) \text{ [Kerr–Newman, future]} \\
& \text{Kerr–Newman metric (charged rotating black hole)}
\end{aligned}$$

New Predictions

New predictions from the Kerr derivation

K35-P1 (Frame dragging rate). The Lense–Thirring precession frequency for a gyroscope at distance $r \gg r_s$ from a CRF rotating source is:

$$\Omega_{\text{LT}} = \frac{\ln(2) J}{D_0 c^2 r^3} = \frac{GJ}{c^2 r^3}$$

This is the standard GR result, now derived from the δ -chain with $G = \ln(2) c^2 / D_0$. No free parameters.

K35-P2 (Maximum spin bound). The CRF information constraint $\varepsilon_0 > 0$ implies:

$$\frac{a}{r_s/2} = \frac{2Jc}{GM^2} \leq 1$$

This is the extremal Kerr bound, derived here as an information-theoretic statement rather than a censorship conjecture.

K35-P3 (Ergosphere radius in CRF units). The equatorial ergosphere radius is $r_{\text{ergo}} = r_s = 2 \ln(2) M / D_0$. For $a > 0$, the ergosphere extends above the horizon: a region exists where no static resolution is possible. Testable via gravitational wave ringdown frequencies of Kerr black holes.

Honest Assessment: What This Derivation Does and Does Not Do

What is derived: The symmetry reduction $SO(3) \rightarrow SO(2)$ from a rotating source is a necessary consequence of the δ -chain when a preferred axis exists. The Kerr metric structure (specifically: Σ , Δ , the $g_{t\phi}$ cross term, and the two horizons) follows from the modified D_{KL} with no new axioms. The limits (Schwarzschild and flat space) are recovered exactly.

What remains hypothesis (Hypothesis class):

- The exact form $D_{KL}^{(Kerr)} = D_{KL}^{(Sch)} \cdot r^2/\Sigma$ is derived from a physical argument (oblate information density) but not from a formal δ -chain computation. This requires the continuum limit of Instability #17 (C(r) exact derivation) first.
- The identification of the off-diagonal $g_{t\phi}$ component from the frame-dragging of R-events is structurally motivated but not formally computed from the R-event density integral.
- Kerr–Newman (charged case, U(1) residual symmetry) is sketched but not derived.

Gap estimate: The derivation closes the Kerr problem at the “structural” level — the same level as the original Schwarzschild derivation before $DKL \propto C^2$ was established. Formal closure requires the continuum-limit field equation at $SO(2)$ symmetry, analogous to C5-Nonlinear. Status: **Hypothesis** (structurally sound, gaps in formal computation).

Status Table (Updated)

Result	Status	Source
$n = 3$ from $SO(3)$	Derived	CRF SO3 Proof
$G = \ln(2) c^2/D_0$	Derived	CRF Gravitational Constant
Schwarzschild metric	Derived	CRF Schwarzschild v2
$SO(3) \rightarrow SO(2)$ from J	Derived	This document
$\Sigma = r^2 + a^2 \cos^2 \theta$	Derived	This document
$\Delta = r^2 - r_s r + a^2$	Derived	This document
Kerr metric (full)	Hypothesis	This document
$g_{t\phi}$ formal derivation	Open	Requires C5-Kerr ODE
Kerr–Newman metric	Open (sketch)	U(1) residual, future
C(r) continuum limit (#17)	Open	Discrete-to-continuum step

Summary

The Kerr metric is derived from the δ -chain via the mechanism:

$$\mathbf{J} \neq 0 \implies \text{SO}(3) \rightarrow \text{SO}(2) \implies D_{\text{KL}} \rightarrow D_{\text{KL}} \cdot \frac{r^2}{\Sigma} \implies \text{Kerr metric}$$

The spin parameter $a = J/(Mc)$ is identified in CRF as the angular momentum of the δ -accumulation pattern around the preferred axis. Frame dragging is the carry of R-events along the rotation axis. The ergosphere is the region where no isotropic resolution is possible. The extremal bound $a \leq r_s/2$ is an information-theoretic constraint.

The derivation is at **Hypothesis** level — structurally complete, pending formal computation of the C5-Kerr field equation. No new axioms were introduced.

The gravity chain now spans: Schwarzschild (non-rotating) $\rightarrow$ Kerr (rotating) $\rightarrow$ Kerr–Newman (rotating + charged, open).